

Relations



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Cartesian Product & Ordered Pairs



- An **ordered pair** of elements is written in the form (a, b) which is distinct from (b, a) unless $a = b$

Definition (Cartesian Product):

For sets $A, B \in U$, the Cartesian product, or crossproduct, of A and B is denoted by $A \times B$

and $A \times B = \{(a, b) \mid a \in A, b \in B\}$.

Example:

Let $A = \{x, y\}$ and $B = \{a, b, c\}$ then

$$A \times B = \{(x, a), (x, b), (x, c), (y, a), (y, b), (y, c)\}$$

$$\& B \times A = \{(a, x), (a, y), (b, x), (b, y), (c, x), (c, y)\}$$

$$A \times A = \{(x, x), (x, y), (y, x), (y, y)\}$$

Certainly $A \times B \neq B \times A$ (in general)

Example

$$X = \{ 1, 2 \}$$

$$Y = \{ a, b \}$$

$$X \times Y = \{(1,a),(1,b),(2,a), (2,b)\}$$

$$Y \times X = \{ (a, 1), (a,2), (b,1), (b,2) \}$$

Note:

- $X \times Y \neq Y \times X$
- $|X \times Y| = |X| \cdot |Y|$

Determine:

$$X \times X = \{(1,1), (1,2), (2,1), (2,2)\} , M = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$Y \times Y =$$

$$A = \{1,2,3,4,5,6\}$$

$$R = \{(a, b) \in R: a \text{ divides } b\}$$

$$R = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,2), (2,4), (2,6), \\ (3,3), (3,6), (4,4), (5,5), (6,6) \end{array} \right\}$$

1	1	1	1	1	1
0	1	0	1	0	1
0	0	1	0	0	1
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1

Relation : Definition



Definition(Relation): For sets $A, B \subseteq U$, any subset of $A \times B$ is called a *relation* from A to B . Any subset of $A \times A$ is called a *binary relation* on A .

If R is a relation from a set A to another set B then $R \subseteq A \times B$

Remarks:

- In general, for finite sets A, B with $|A|=m$ and $|B|=n$, there are 2^{mn} relations from A to B , including the empty relation as well as the relation $A \times B$ itself.
- The domain of a relation R is the set of all first elements of the ordered pairs which belong to R , and the range of R is the set of second elements.

Example:

- Let $R = \{ (1,a),(2,b),(1,c) \}$ then $\text{dom } R = \{ 1,2 \}$ and $\text{range } R = \{ a,b,c \}$
- Let $A = \mathbb{Z}^+$, a binary relation, R , on A , $\{ (x,y) \mid x < y \}$
 $(1,2), (7,11)$ is in R , but $(2,2), (3,2)$ is not in R
or $1R2, 7R11$ (infix notation)
- Let $X = \{ 2, 3, 4 \}$ and $Y = \{ 3, 4, 5, 6, 7 \}$
If we define a relation R from X to Y by
 $(x,y) \in R$ if x divides y (with zero remainder)
We obtain:
 $R = \{ (2,4), (2,6), (3,3), (3,6), (4,4) \}$
Domain of R is $\{ 2, 3, 4 \}$; Range of R is $\{ 3, 4, 6 \}$

Another Example:



- Let R be the relation on $X = \{1,2,3,4\}$ defined by $(x,y) \in R$ if $x \leq y$, $x, y \in X$.
- $R = \{ (1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4) \}$
- The domain and Range of R are both equal to X

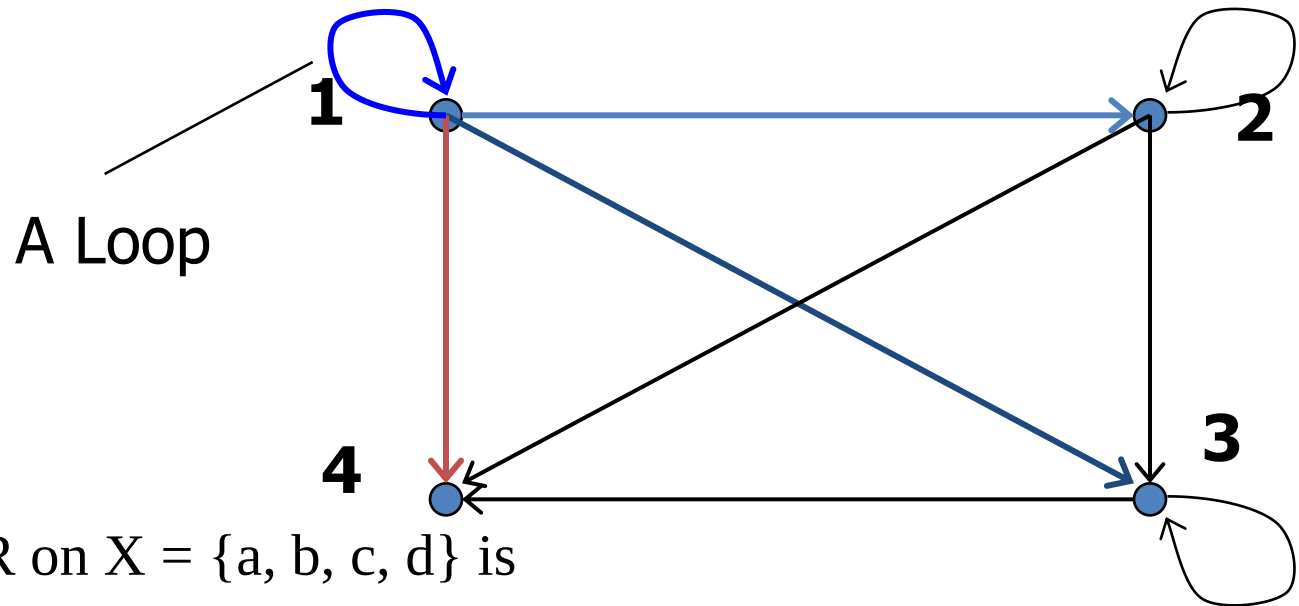
Representing Relations using a Digraph



- Digraph – directed graph
- Components –
 - Vertices or Nodes, drawn as dots
 - Edges - Directed edges i.e. lines between vertices with arrows showing a direction between the vertices)
- Vertices represent elements of X
- Directed Edge represent the members of the relation R

Example

Let $R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4)\}$ be a relation on $A = \{1,2,3,4\}$



Exercise:

- The Relation R on $X = \{a, b, c, d\}$ is

$$R = \{(a,a), (b,c), (c,b), (d,d)\}$$

What would the digraph to represent this relation look like?

Representing Relations using a Matrix

Let $A = \{a_1, a_2, \dots, a_m\}$ and $B = \{b_1, b_2, \dots, b_n\}$

Let R be relation from A to B . Then R is represented by a matrix $M_R = [m_{ij}]_{m \times n}$ where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$$

The matrix M_R is called the matrix of R .

Example : Let $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$

Let $R = \{(1, y), (2, z), (3, x), (3, y), (4, x), (4, z)\}$ be a relation from A to B .

Then M_R is given by

	x	y	z
1	0	1	0
2	0	0	1
3	1	1	0
4	1	0	1

$$\therefore M_R = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

Types of Relations:

- Reflexive
- Irreflexive
- Symmetric
- Antisymmetric
- Transitive

Reflexive and Irreflexive

- A relation R on a set A is called **reflexive** if $(a, a) \in R$ for every $a \in A$

e.g.

$R = \{ (1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4) \}$

is Reflexive. [$A = \{1,2,3,4\}$] while

$R^* = \{ (1,1), (1,2), (1,3), (1,4), (2,3), (2,4), (3,3), (3,4) \}$

is not a reflexive relation on A .

- A relation R on a set A is called **irreflexive** if (a,a) does not belong to R for any $a \in A$.

$R' = \{ (1,2), (1,3), (1,4), (2,3), (2,4), (3,4) \}$ *is irreflexive relation on A .*

Symmetric and Anti-symmetric

- A relation R on a set A , is called **symmetric** if whenever $(a,b) \in R$, then $(b,a) \in R$

Example:

$R_2 = \{(a,a), (b,c), (c,b), (d,d)\}$ is symmetric.

Is R_2 also Reflexive?

- A relation R on a set A is called *antisymmetric* if whenever (a, b) and $(b, a) \in R$ then $a=b$.

Example:

$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$
is Antisymmetric as $(2,1)$, $(3,1)$ and $(4,2)$ are not in the relation R .

- Note that this relation is not symmetric but it is antisymmetric.

Antisymmetric (continued)



- Is $R_2 = \{(a, a), (b, c), (c, b), (d, d)\}$ is antisymmetric?
No. Both (b, c) and (c, b) are in R .
- If a relation R on X has no members of the form (x, y) with $x \neq y$, then R is antisymmetric. **WHY?**
- **Let** $R = \{(a, a), (b, b), (c, c)\}$
then R is antisymmetric
Is R also Symmetric? Yes.

NB. "Antisymmetric" is not the same as "NOT symmetric"

Transitivity

- A relation R on set A is called **transitive** if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$

Example:

Let $R = \{ (1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4) \}$ is transitive but

$R = \{ (a, a), (b, c), (c, b), (d, d) \}$ is **not** transitive. WHY?

Inverse of a relation

- Let R be a relation from A to B . The **inverse** of R , denoted R^{-1} , is the relation from B to A defined by

$$R^{-1} = \{(y, x) | (x, y) \in R\}$$

- $X = \{2, 3, 4\}$ and $Y = \{3, 4, 5, 6, 7\}$, a relation R is defined by

$(x, y) \in R$ if x divides y (with zero remainder)

$$R = \{(2, 4), (2, 6), (3, 3), (3, 6), (4, 4)\}$$

$$\text{Then } R^{-1} = \{(4, 2), (6, 2), (3, 3), (6, 3), (4, 4)\}$$

The composition of two relations

Let R be a relation from A to B and S be a relation from B to C . The **composition** of **R and S** , denoted by **$S \circ R$** , is the relation from A to C defined by

$$S \circ R = \{(a, c) \mid (a, b) \in R \text{ and } (b, c) \in S \text{ for some } b \in B\}$$

Example: Let R is the relation from $\{1,2,3\}$ to $\{1,2,3,4\}$ with $R = \{(1,1), (1,4), (2,3), (3,1), (3,4)\}$ and S is the relation from $\{1,2,3,4\}$ to $\{0,1,2\}$ with $S = \{(1,0), (2,0), (3,1), (3,2), (4,1)\}$. Find $S \circ R$.

Solution: $S \circ R = \{(1,0), (1,1), (2,1), (2,2), (3,0), (3,1)\}$.

EXAMPLE

Let R and S be binary relations on $\{1, 2, 3, 4\}$ such that

$$R = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$$

$$S = \{(1, 2), (2, 1), (3, 4), (4, 3)\}$$

$$S \circ R = \{(1, 1), (2, 4), (3, 3), (4, 2)\}$$

$$R \circ S = \{(1, 3), (2, 2), (3, 1), (4, 4)\}$$

Find $R \circ R$ and $R \circ S \circ R$

IF there is n elements in a set then

- Number of **reflexive** relation = $2^{n(n-1)}$
- Number of **symmetric** relation = $2^{n(n+1)/2}$
- Number of **anti-symmetric** relation = $2^n \cdot 3^{n(n-1)/2}$
- Number of **asymmetric** relation = $3^{n(n-1)/2}$
- Number of **irreflexive** relation = $2^{n(n-1)}$
- Number of **reflexive and symmetric** relation = $2^{n(n-1)/2}$
- Number of **neither reflexive nor irreflexive** relation =
$$2^{n^2} - 2 \cdot 2^{n(n-1)}$$

Partial Order relation: Introduction



- An equivalence relation is a relation that is reflexive, symmetric, and transitive
- A **partial ordering (or partial order)** is a relation that is **reflexive**, **antisymmetric**, and **transitive**
 - Recall that antisymmetric means that if (a,b) and $(b,a) \in R$, then $b = a$
 - Thus, (a,a) is allowed to be in R
 - But since it's reflexive, all possible (a,a) must be in R
- A set **S** with a partial ordering **R** is called a **partially ordered set, or poset**
 - Denoted by (S,R)

Partial ordering examples

- Show that $\geq$ is a partial order on the set of integers
 - It is reflexive: $a \geq a$ for all $a \in \mathbf{Z}$
 - It is antisymmetric: if $a \geq b$ and $b \geq a$ then $b = a$
 - It is transitive: if $a \geq b$ and $b \geq c$, then $a \geq c$
- Note that $\geq$ is the partial ordering on the set of integers
- $(\mathbf{Z}, \geq)$ is the partially ordered set, or poset.
- Show that the inclusion relation $\subseteq$ is a partial ordering relation on the power set of a set S .
(Exercise!) i.e. $(\mathbf{P}(S), \subseteq)$ is a poset.

Partial Orderings

- A partial ordering (or partial order) is a relation that is reflexive, *antisymmetric*, and transitive
 - Recall that antisymmetric means that if $(a,b) \in R$, then $(b,a) \notin R$ unless $b = a$
 - Thus, (a,a) is allowed to be in R
 - But since it's reflexive, all possible (a,a) must be in R

Partially Ordered Set (POSET)



- A relation R on a set S is called a *partial ordering* or *partial order* if it is *reflexive*, *antisymmetric*, and *transitive*. A set S together with a partial ordering R is called a *partially ordered set*, or *poset*, and is denoted by (S, R)

Comparable / Incomparable

- The elements a and b of a poset $(S, \leq)$ are called *comparable* if either $a \leq b$ or $b \leq a$. When a and b are elements of S such that neither $a \leq b$ nor $b \leq a$, a and b are called *incomparable*.

Example

- Consider the power set of $\{a, b, c\}$ and the subset relation.
 $(P(\{a, b, c\}), \subseteq)$

$$\{a, c\} \not\subseteq \{a, b\} \text{ and } \{a, b\} \not\subseteq \{a, c\}$$

So, $\{a, c\}$ and $\{a, b\}$ are *incomparable*

- In the poset $(\mathbf{Z}^+, \leq)$, are the integers 3 and 9 comparable?
 - Yes, as $3 \leq 9$
- Are 7 and 5 comparable?
 - Yes, as $5 \leq 7$
- As all pairs of elements in $\mathbf{Z}^+$ are comparable, the poset $(\mathbf{Z}^+, \leq)$ is a total order
 - a.k.a. totally ordered poset, linear order, or chain

- In the poset $(\mathbf{Z}^+, |)$ with “divides” operator $|$, are the integers 3 and 9 comparable?
 - Yes, as $3 | 9$
- Are 7 and 5 comparable?
 - No, as $7 \nmid 5$ and $5 \nmid 7$
- Thus, as there are pairs of elements in $\mathbf{Z}^+$ that are not comparable, the poset $(\mathbf{Z}^+, |)$ is a partial order. It is not a chain.